

Yukawa potential

Find σ_{total} for the Yukawa potential

$$V(r) = -\frac{V_0 \exp(-\mu r)}{\mu r}$$

using the Born approximation

$$f(\theta) = \frac{m}{2\pi\hbar^2} \int \exp(i\mathbf{k} \cdot \mathbf{r}) V(\mathbf{r}) d\mathbf{r}$$

where $\mathbf{k}$ is momentum transfer

$$\mathbf{k} = \mathbf{k}_i - \mathbf{k}_f$$

and

$$k^2 = |\mathbf{k}|^2 = \frac{4mE(1 - \cos\theta)}{\hbar^2}$$

Convert to polar coordinates.

$$f(\theta) = \frac{m}{2\pi\hbar^2} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp(ikr \cos\theta) V(r, \theta, \phi) r^2 \sin\theta dr d\theta d\phi$$

Substitute

$$V(r, \theta, \phi) = -\frac{V_0 \exp(-\mu r)}{\mu r}$$

to obtain

$$f(\theta) = -\frac{mV_0}{2\pi\hbar^2\mu} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp(ikr \cos\theta - \mu r) r \sin\theta dr d\theta d\phi$$

Integrate over ϕ (multiply by 2π).

$$f(\theta) = -\frac{mV_0}{\hbar^2\mu} \int_0^\infty \int_0^\pi \exp(ikr \cos\theta - \mu r) r \sin\theta dr d\theta$$

Let $y = \cos\theta$ and $dy = -\sin\theta d\theta$. The minus sign in dy is canceled by interchanging integration limits $\cos(0) = 1$ and $\cos(\pi) = -1$.

$$f(\theta) = -\frac{mV_0}{\hbar^2\mu} \int_0^\infty \int_{-1}^1 \exp(ikry - \mu r) r dr dy$$

Solve the integral over y .

$$f(\theta) = -\frac{mV_0}{\hbar^2\mu} \int_0^\infty \left[\frac{1}{ikr} \exp(ikry - \mu r) \right]_{y=-1}^{y=1} r dr$$

Cancel r and evaluate the limits.

$$f(\theta) = -\frac{mV_0}{\hbar^2\mu} \frac{1}{ik} \int_0^\infty [\exp(ikr - \mu r) - \exp(-ikr - \mu r)] dr$$

Solve the integral over r .

$$f(\theta) = -\frac{mV_0}{\hbar^2\mu} \frac{1}{ik} \left[\frac{1}{ik - \mu} \exp(ikr - \mu r) + \frac{1}{ik + \mu} \exp(-ikr - \mu r) \right]_{r=0}^{r=\infty}$$

Evaluate the limits. The exponentials vanish at the upper limit.

$$f(\theta) = -\frac{mV_0}{\hbar^2\mu} \frac{1}{ik} \left[-\frac{1}{ik - \mu} - \frac{1}{ik + \mu} \right] = -\frac{2mV_0}{\hbar^2\mu} \frac{1}{k^2 + \mu^2} \quad (1)$$

Substitute for k^2 .

$$f(\theta) = -\frac{2mV_0}{\mu} \frac{1}{4mE(1 - \cos \theta) + \mu^2\hbar^2} \quad (2)$$

Hence

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2 = \left(\frac{2mV_0}{\mu} \right)^2 \frac{1}{[4mE(1 - \cos \theta) + \mu^2\hbar^2]^2}$$

For the total cross section we have

$$\sigma_{\text{total}} = \int \frac{d\sigma}{d\Omega} d\Omega = \left(\frac{2mV_0}{\mu} \right)^2 \int_0^\pi \int_0^{2\pi} \frac{1}{[4mE(1 - \cos \theta) + \mu^2\hbar^2]^2} \sin \theta d\theta d\phi$$

Integrate over ϕ (multiply by 2π).

$$\sigma_{\text{total}} = 2\pi \left(\frac{2mV_0}{\mu} \right)^2 \int_0^\pi \frac{1}{[4mE(1 - \cos \theta) + \mu^2\hbar^2]^2} \sin \theta d\theta$$

Let $y = 1 - \cos \theta$ and $dy = \sin \theta d\theta$. The limits transform as $1 - \cos(0) = 0$ and $1 - \cos(\pi) = 2$.

$$\sigma_{\text{total}} = 2\pi \left(\frac{2mV_0}{\mu} \right)^2 \int_0^2 \frac{1}{[4mEy + \mu^2\hbar^2]^2} dy$$

Solve the integral.

$$\sigma_{\text{total}} = 2\pi \left(\frac{2mV_0}{\mu} \right)^2 \left[-\frac{1}{4mE(4mEy + \mu^2\hbar^2)} \right]_{y=0}^{y=2}$$

Evaluate the limits.

$$\sigma_{\text{total}} = 2\pi \left(\frac{2mV_0}{\mu} \right)^2 \frac{2}{8mE\mu^2\hbar^2 + \mu^4\hbar^4} \quad (3)$$

For the dimensions of σ_{total} we have

$$\begin{aligned} mV_0 &\propto \text{mass} \times \text{energy} = \text{momentum}^2 \\ mE &\propto \text{mass} \times \text{energy} = \text{momentum}^2 \\ \mu\hbar &\propto \text{length}^{-1} \times \text{energy} \times \text{time} = \text{momentum} \end{aligned}$$

Hence

$$\begin{aligned}(mV_0)^2 &\propto \text{momentum}^4 \\ mE\mu^2\hbar^2 &\propto \text{momentum}^4 \\ \mu^4\hbar^4 &\propto \text{momentum}^4\end{aligned}$$

The momentum dimensions cancel leaving units of area.

$$\sigma_{\text{total}} \propto \frac{1}{\mu^2} \propto \text{length}^2$$

See exercise 10.7 of *Quantum Mechanics* by Richard Fitzpatrick.